

Extracted Features and Explanation of Data

The dataset used is the NASA Battery Data Set (<https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>). In table 1, recordings were collected from experiments on lithium ion batteries and then placed into MATLAB files.

During the experiment, 29 lithium batteries were tested under charge, discharge, and impedance operation types at different temperatures. The charging operation was carried out at a fixed current of 1.5 Amps (A) until the voltage reached 4.2 Volts (V), the impedance operation was measured using an EIS sweep from 0.1 Hz to 5 kHz to track internal parameter changes during aging, and the discharging operation was carried out from 4.2 V until the voltage reached specific voltage levels (2.7 V ,2.5 V, 2.2 V and 2.0 V).

The dataset contains each battery operation (charge, discharge, or impedance), with ambient temperature, timestamps, and detailed measurement fields such as measured voltage, current, temperature, charger/load voltage and current, time vectors, discharge capacity, and impedance-related values including sense current, battery current, current ratio, battery impedance, rectified impedance and estimated resistances (Re and Rct).

Table 1. Initial fields and definitions from the researchers that conducted the battery experiments [33].

Fields	Definitions
Cycle	Represents each time an operation type was recorded and not the actual cycle (full discharge and recharge).
Type	Operation type, this can be charge, discharge or impedance.
Time	The date and time of the start of the operation type.

Ambient_temperature	The ambient temperature (degree C)
Charge Fields	
Voltage_measured	Battery voltage (V)
Current_measured	Battery output current (A)
Temperature_measured	Battery temperature
Current_charge	Current measured (A) at the charger.
Voltage_charge	Voltage measured (V) at the charger.
Time	Time taken for operation type.
Discharge fields	
Voltage_measured	Battery voltage (V)
Current_measured	Battery output current (A)
Temperature_measured	Battery temperature
Current_charge	Current measured (A) at the load.
Voltage_charge	Voltage measured (V) at the load.
Time	Time taken for operation type.
Capacity	Battery capacity (Amp/hr) for discharge operation.
Impedance fields	
Sense_current	Current in sense branch (A)

Battery_current	Current in battery branch (A)
Current_Ratio	Ratio of the above currents
Battery_impedance	Battery impedance (Ohms) computed from raw data
Rectified_impedance	Calibrated and smoothed battery impedance (Ohms)
Re	Estimated electrolyte resistance (Ohms)
Rct	Estimated charge transfer resistance (Ohms)

A data collection process, implemented in Python, was used to extract the battery data from the original MATLAB files, transform it into useful features, and convert it into a dataframe and a CSV file.

Refer to table 2, Seventeen features were extracted from the battery data (which was in MATLAB format). An investigation was conducted to better understand and extract features from the data, particularly because the original battery dataset was not in a table format, and most features were expressed as results over time for each cycle phase (discharge, charge, impedance). For example, the discharge "Voltage_measured" had multiple voltage measurements per cycle across one test. Essentially, the researchers were measuring the voltage drop as the battery was being discharged. As a result, the discharge voltage had to be aggregated using its maximum (voltage before discharging) and minimum (voltage after fully discharging). A similar aggregation process was implemented on various other features.

No impedance values were included in the seventeen features because the researchers did not track the impedance throughout each cycle. Since impedance values were inconsistently measured, this would have resulted in empty values across many data points. Only the measurements pertaining to the battery were included, as it is assumed that only battery-related information would be obtainable (please refer to Section 1.3, Purpose). As a result, measurements relating to the charger or load were not extracted.

Although the discharge and charge were recorded separately in the original MATLAB file, a Python script was used to "normalize" both the discharge and charge phases onto

the same data points. This allowed both sets of features to be used when predicting the State of Health (SOH), which is the target variable.

Table 2. Extracted features from original matlab file (after python transformation)

Features	Description
Battery_measurement_time	The timestamp represents the start of each charge cycle.
Battery name (categorical)	The battery number associated with each battery used in the experiments. NB, not the actual name of the battery type.
Cycle (derived feature)	The number of cycles. In this experiment, a battery full cycle occurs when a full battery (100%) is discharged and then recharged back to 100% capacity. Each time this discharging and charging occurs then 1 cycle has occurred.
Charge_time_measured	The time taken to charge the battery.
Charge_min_Voltage	The minimum voltage measured during the charging phase.
Charge_max_Voltage	The maximum voltage measured during the charging phase.
Charge_min_Current_measured	The minimum current measured during the charging phase.
Charge_max_Current_measured	The maximum current measured during the charging phase.
Charge_Temp_measured	The average temperature during the charging phase.
Discharge_time_measured	The time taken to discharge the battery.
Discharge_min_Voltage_measured	The minimum voltage measured during the discharging phase.
Discharge_max_Voltage_measured	The maximum voltage measured during the discharging phase.
Discharge_min_Current_measured	The minimum current measured during the discharging phase.
Discharge_max_Current_measured	The maximum current measured during the discharging phase.
Discharge_Temp_measured	The average temperature during the discharging phase.
Capacity	The capacity value after discharging
SOH (derived feature)	State of Health. This is calculated as the measured capacity at a

	specific cycle, divided by the capacity measured during the first cycle.
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Please see the voltage measurements (during the discharging phase) after the matlab data files were transformed into a python array. The voltage starts at 4.1915V (Maximum voltage and is the feature “Discharge_max_Voltage_measured”) and then slowly drops throughout the discharging phase.

```
[17]
✓ Os (battery["cycle"][0,0][0,1]["data"][0,0]["Voltage_measured"])
▼
... array([[4.19149181, 4.19074907, 3.97487891, 3.95171671, 3.93435249,
3.92005844, 3.90790351, 3.89703572, 3.88747658, 3.87895859,
3.87101598, 3.86347968, 3.85666051, 3.85001358, 3.84384191,
3.83768798, 3.83193825, 3.82641444, 3.82109573, 3.81589666,
3.81069682, 3.80583738, 3.80112163, 3.79629313, 3.79144891,
3.78662469, 3.78206292, 3.7771754 , 3.77240629, 3.76793746,
3.7632293 , 3.75851149, 3.75421321, 3.7496292 , 3.74516886,
3.74100501, 3.73669983, 3.73248271, 3.72835255, 3.72453324,
3.72065103, 3.71640901, 3.71279184, 3.70842761, 3.70479177,
3.70068258, 3.69675771, 3.69303933, 3.6891768 , 3.68521448,
3.68149374, 3.67779221, 3.67406181, 3.67030176, 3.66673391,
3.66299085, 3.65966439, 3.65589414, 3.65232651, 3.64876348,
3.64554686, 3.64207625, 3.63846406, 3.63507607, 3.63166482,
3.62857854, 3.6250126 , 3.62158223, 3.61846125, 3.61506725,
3.61188181, 3.60857868, 3.60553494, 3.6021859 , 3.59885715,
3.59576709, 3.59301666, 3.58971009, 3.5866522 , 3.58344204,
3.58048534, 3.57725048, 3.57436883, 3.57128513, 3.56872067,
3.56555467, 3.56259541, 3.55978447, 3.55689375, 3.55397359,
3.55119783, 3.5485156 , 3.54500583, 3.54311072, 3.54029356,
3.53740101, 3.53518704, 3.53250094, 3.52990334, 3.52754753,
3.52501551, 3.52260336, 3.52011278, 3.51796895, 3.51538625,
3.51328369, 3.51113986, 3.50880484, 3.50689115, 3.5047116 ,
3.50278637, 3.50064119, 3.49884032, 3.49651009, 3.49476208,
3.49264984, 3.49090381, 3.48884863, 3.48686929, 3.48499587,
3.48317381, 3.48099635, 3.47932883, 3.47721288, 3.47531054,
3.47344329, 3.4711879 , 3.46915804, 3.46715799, 3.46509799,
3.46319749, 3.46076571, 3.45852954, 3.45637595, 3.45382105,
3.45159071, 3.44934342, 3.44691178, 3.44433708, 3.44186105,
3.43940027, 3.43636103, 3.43367081, 3.43080658, 3.4278209 ,
3.42484406, 3.42175727, 3.41874632, 3.41545781, 3.41177601,
3.40829397, 3.40435779, 3.40023624, 3.39613072, 3.39181338,
3.38704829, 3.38203092, 3.37654809, 3.37045168, 3.36378703,
3.35685219, 3.34867749, 3.33959009 , 3.32963513, 3.31820701,
3.30494144, 3.29037737, 3.27371569, 3.25451646, 3.23310264,
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Discharge Voltage measured